Hybrid Texture-and-Deep-Feature Fusion for Melanoma Detection

(2022)

Problem

Detecting melanoma from dermoscopic images under limited labeled data.

Method

Combines Gabor filters, Local Binary Patterns (LBP), and Histogram of Oriented Gradients (HOG) with a Random Forest classifier, benchmarked against a deep CNN baseline.

Dataset

HAM10000 dermoscopic image dataset.

Evaluation

Precision, recall, and F1-score.

Results

The hand-crafted feature fusion reached F1 of 0.83, close to the CNN baseline's 0.86, while training in a fraction of the time.

Limitations

Hand-crafted features remain competitive on small data but do not generalize across imaging devices from different manufacturers.